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Past, present and future of design ethics

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Abstract: With the ever-more present climate crisis, resource shortage and the expansion of AI, the core question of the ethical aspects of our development of new products and solutions becomes unavoidable. But which topics are commonly discussed in design ethics literature and how has it evolved? To answer this question and to understand the existing underpinnings and foundation of ethics in design, we adopted a structured literature review searching literature across 42 renowned scientific design journals. A systematic search revealed 1177 academic publications relating to the topic. After filtering and reviewing the titles, abstracts and keywords of these publications, a total of 121 sources were singled out as qualified to constitute the foundation of this review. From these, 8 central themes in past and present design ethics research were identified: Design processes and practices, design education, participatory/co-creation, responsible design, social design, sustainability, technology, and human-centred design of which the three predominant themes are related to questions of the designers' role and actions (design processes and practices), the effects and consequences of new developments (technology), and the common objectives of addressing the urgency of ecological imbalance (sustainability). This opens a discussion of future research, what are we missing?

Keywords: Design ethics, Structured Literature Review, Technology, Sustainability

1. Introduction

With the ever-more resent climate crisis, resource scarcity, and the expansion of AI, the core question of the ethical aspects of our development of new products and solutions becomes unavoidable. Contemporary discussions in the field of design ethics typically revolve around the inherent ethical properties of products as sophisticated technologies increasingly integrate with everyday products. But how do we engage in this discussion? First, we need to understand the existing underpinnings and foundation of ethics within design.

Ethical design has been framed as a moral reflection on creative actions undertaken in the design process, as well as a process that both transforms and extends potentiality (Buwert, 2017). While design has been described as: "... the most powerful tool with which man shapes his tools and environments (and, by extension, society and himself)." (Papanek,



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2019, s. IX) researchers argue that designers then assume the role of ‘material ethicists’. Design ethics can thus be defined as both a natural extension of product design and, conversely, product design as an inherently ethical activity (Birkler, 2019; Haidamous, 2017).

The topic of technology and ethics at the level of industrial design have prior been visited by authors such as Victor Papanek (1972) and Tomas Maldonado (1972), while other contributors on the topic include the works by Douglas and Isherwood (1979), Norman (1988), Simondon (2011), and Manzini (1990).

Evidently, within design research, ethics has a long history, but despite this, the area remains underdeveloped (Chan, 2018; Fry, 2004; Dindler, et. al., 2022). In recent years Chan (2018) has called for a revitalization of ethics in design which Fry (2004) has also argued remains a “stranded debate” because of the still infant cultures of design.

Based on these observations, we find it advantageous to develop a scholarly body of literature that can provide potentially new research paths to the topic of design ethics. Our goal is to offer a straightforward and accessible overview of the insights gained from this study, encouraging widespread participation in scientific discussions on the topic of design ethics.

This article will lay down the theoretical fundament needed to explore contemporary design ethics focusing on core design literature. For now, we formulate our research questions to form an overview of the topic at hand. For this purpose, design ethics will be explored through a structured literature review conducted within 42 renowned design journals in which design is a common field of interest – this list has been constructed from a study on the topic of circular economy within industrial design (van Dam et.al., 2020).

The research question is the following: Which topics are commonly researched in (industrial) design ethics literature and how has it evolved?

2. What is design ethics? And why is it important?

In a philosophical context ethics is concerned with what constitutes a “good”, “harmonious”, and/or “virtuous” life as well as how one should live. The latter refers to a universal set of rules or norms (also referred to as a universal “code of conduct”) to which we conform when among the public (Parsons, 2016). In society, rules will operate on different levels, for example by scope and by consequences that follow disobeying them. How such rules of society are communicated can also be very different - some may be explicitly written (as in the law) while others may be deeply embedded into the shared societal mindsets around the world (Ethics). An example of the *less serious versions of the above examples* of ethics is etiquette, while another example, related to rules in the law, could be the written rules of a game (Birkler, 2019).

“...design is by nature an activity which extends and transforms *potentiality* and that therefore, because of this, it is always an *ethical* activity. This foundational ethicality does not guarantee that design will always be good, but rather that it always possesses within itself the simultaneous potentiality for both good and evil.” (Buwert, 2017, p S4459).

Ethics is about what is uniformly believed to be “good”, “harmonious” and/or “virtuous”.

Morals is about conforming to these beliefs.

Design is about the way in which we create and give form to these beliefs.

(d’Anjou, 2009; Birkler, 2019).

“With **design** we can solve problems, form, and change our surroundings, as well as grant specific needs to others. However, with **ethics**, we may ask about if what was done was the right thing to do, and whether it was done in the right way.” (Birkler, 2019, p. 9¹).

In the quote, Birkler (2019) illustrates how ethics can be viewed as a natural extension of design arguing that design and ethics are framed through a moral reflection on the creative actions that are taken in the design process. Moreover, as design is a projective practice, outwards to its context and its human benefactors, ethical thinking and ethical assessment are natural to consider. (Fel-

ton et. al., 2013).

3. Methodology

This study adopts a structured literature review (Massaro et. al., 2016) intending to cover a large scholarly body of literature on the parameters of width and depth. The *width* is obtained through exhaustive data search and collection while *depth* is added through a validity construct based on assessment criteria.

A rigorous method such as the structured literature review (“SLR” in short) enables researchers to uncover general traits and patterns within a larger body of literature on any scientific topic. Specific steps are usually taken to build a logical review structure that is both transparent and replicable. Based on this, we first developed a protocol specifying the approach of data collection, processing, and evaluation, including information such as search engines, source specifications, and inclusion/exclusion criteria for coding into the pre-structured analytical framework. The research *protocol* in *Tables 1* and *2* ensures that the collection and extraction of data is performed following our pre-planned structure.

Table 1: The SLR Protocol: Literature search-scope

Parameters	Specifications
Search engines	- Scopus (main search engine) - Primo search engine (AUB Library) & Google Scholar (supporting search engines)

¹Translated from the original language (Danish).

Data source Journals Top 42 rated journals in design research covering the field of industrial design (van Dam et. al., 2020)	Applied ergonomics CoDesign Design And Culture Design Issues Design studies Digital Creativity Ergonomics in design Interacting with Computers International Journal of Art and Design Education Int Journal of Design Int Journal of Product Development Int Journal of technology and design education Int Journal on Interactive Design and Manufacturing Journal of Design History Journal of Design Research Journal of Interior Design Journal of Material Culture Journal of Mechanical Design [transactions of the ASME] She Ji The Design Journal Visual Communication Creativity and Innovation Management	CAD Computer-Aided Design Design Science Ergonomics Empirical Studies of the Arts Human-Computer Interaction Human Factors Information Design Journal Journal of Product Innovation Management Journal on Multimodal User Interfaces Materials and Design/No art Personal and Ubiquitous Computing Research in Engineering Design The Senses & Society Design Management Journal* Design Philosophy Papers Design Research Quarterly Journal of Engineering Design int Journal of Sustainable Design The Journal of Sustainable Product Design Visible Language
Timeframe	1970 to 2023 (cutoff date: 1/9-2023)	

The literature search encompasses 42 of the most esteemed journals in design research with a focus on industrial design (van Damet. al., 2020). The primary data search will be conducted using Scopus. Support for this process will be provided by the Primo/AUB Library (Aalborg University Library) search engine and Google Scholar, enabling the collection of PDFs that may not be assessable through Scopus, as well as conducting (meta)data searches for journals not included in the Scopus library.

The protocol is designed to direct a controlled reduction of the search output while filtering out non-relevant sources. While we aim to generate a versatile and representative dataset the constricting nature of working within specific parameters of data (and types of data) will unavoidably lead to working with a narrower data pool altogether. Because of this, we chose few, but broad keywords in our search to capture as much data as possible. Another thing to be aware of when working with different journals is that some may be *discontinued* or only actively published over specific periods. Some may publish several issues a month while others publish every quarter, which ultimately is a deciding factor in how many results are realistically possible to receive from the search. Because of this, we kept a record of the active

periods of the journal as well as the total number of publications recorded from each journal. Comparing our search hits to the recorded information on each journal will help us assess whether the search results seem out of proportion to the existing number of publications in each journal. Table 2: The SLR Protocol: Literature search terms and rules and the quality assessment criteria.

We chose the search metric as displayed below and *added a maximum of five Journal-IDs at a time* for each segment of executed search commands.

//TITLE-ABS-KEY (design AND ethics OR “design ethics”) OR (“design” AND “ethical” OR “ethical design”) (input...) (Source-ID (“-Journal ID2 no.-“)

By consulting the pre-recorded publication ratings on a select few journals at a time, we will be able to easier spot inconsistencies or any flawed commands in a structured manner, while keeping a track record of literature and/or meta-data checks in other databases/through AUB-library support. Initially, we landed on a total of 1177 hits using combinations of the two constants (A, B.1, B.2).:

TITLE-ABS-KEY(A (variations) AND B1. (variations) OR “AB.1”) OR (A (variations) AND B2. (variations) OR “AB.2”)

An additional keyword search was performed, and the results were collectively screened (filtering the correct format, removing duplicates, and removing any journals not listed – if a journal has a similar title to any of the 42 on the list)

TITLE-ABS-KEY(A (variations) AND B1. (variations) OR “AB.1”) OR (A (variations) AND B2. (variations) OR “AB.2”) OR ((“AB.1”/“AB.2”) AND Add. (variations))

When we reached 568 sources an additional screening focusing on headline/keyword eligibility was conducted while filter options and metrics got a second check. When we got the total down to 360 results a process with more manual filtering was initiated. In team meetings abstracts were read, compared, and collapsed into groups of similar themes. These clusters were then assessed according to the four priority assessment criteria (Protocol Table 1) which was also discussed within the research group. Finally, *121 articles were imported* to the QDAS program Nvivo for further analysis.

One author was responsible for the technical planning, collection, meta-data recording, and assessment/sorting described within this chapter. To secure a thorough downsizing of the manual parts, all researchers within the team took turns reading through and discussing pre-sorted piles of abstracts.

² The (“-Journal ID no.-“) looks like this when filled out: (SRCID(13929) OR SRCID(144972) OR SRCID(19700201037) OR SRCID(19700200907) (...)

4. Data analysis

4.1 Coded themes

The themes below are based on nodes (containers of coded text passages and other information) from 121 articles which have been analyzed in the qualitative data analysis software (QDAS) program NVivo (version R.1.1). The schema provides definitions for selected themes and/or terms coded under the thematic nodes. The third column contains references to articles that are representative of the thematic nodes.

Table 3 Central themes and subthemes generated from analyzing the scholarly body of design ethics literature.

Themes & sub-themes in Design Ethics literature	Definition(s)	Representative sources
1. Design processes & Practices (48) 1.1 Decision making (21) - Consequences /local effects vs. global/societal effects - Agency & Intentionality - Rational choice theory (Preconventional/Conventional) *	Agency & Intentionality: - <i>Intentionality</i>: where intentions are acts influenced by prior intentions, intentionality is intention in action or experience of acting (Sorabji, 1991). - <i>Agency</i> refers to a person's ability to pursue and realize goals that he or she values and has reason to value. The opposite of a person with agency is forced, oppressed, or passive (van den Hoven, 2015, p. 224; Alkire 2008, p.3) - Having <i>ethical agency</i> as a designer is about having the capability to identify and make the "right" decisions (what resonates with one's moral philosophy) and to act on these by making choices that bring change in a positive way (van der Velden, 2009) - <i>Rational choice theory</i> is morally holding up the premise that people who face a decision choose among fully reasoned, exhaustively considered alternatives, to make the choice they think will serve them best. (Dekker, 2017, p. 554) <i>*Preconventional (self-interest) / Conventional (social norms)</i>	1.1 Van Gorp (2007) Matei (2018) Dekker (2017) Van Der Velden (2009)
1.2 Research methods and tools (14) - Abduction - Reasoning - Evolutionary product development - NPD New product development	- <i>Abduction</i> can be conceptualized as making guesses. In a theory building process there is a need to make guesses when a researcher makes observations which are surprising in that they depart from existing theory. Other researchers have used abduction to describe and understand theory building (Phillipsen, 2018, p. 45) - <i>Reasoning</i> is the ability to assess things rationally by applying logic. Defined as "the drawing of inferences or conclusions through the use of reason" (Merriam-Webster 1, 2024) - <i>Evolutionary Product Development (EPD)</i> is a design approach based on the observation that products typically go through a series of phases after their initial market introduction (Ehlhardt, 2013, p. 8). - <i>New product development (NPD)</i> Refers to the process of delivering a new product, service, idea, or technology to the market. The process involves several stages, from ideation and concept development to market research, product launch, and evaluation (SC L2, 2024)	1.2 Faludi et. El., (2020) Steen (2013) Loyd (2009) McDonald et. al. (2019) Koo (2016) Eger and Drukker (2012)
1.3 Theoretical/ Conceptual frameworks (11)	- <i>CSV Creating shared value</i>	1.3

- CSV creating shared value. - Positive design/design for wellbeing - Potentiality theory	Creating shared value (CSV) is a concept developed by Michael E. Porter and Mark R. Kramer in 2011 focusing on the relationships between economic and societal progress. Creating shared value is a framework for creating economic value while simultaneously addressing societal needs and challenges (HEC L3, 2024; Kramer, 2011). - According to Desmet and Pohlmeier (2013) <i>Positive design</i> is an umbrella term for design “positive design” as an umbrella term for all forms of design, design research and design intention in which explicit attention is paid to the effects of design on the subjective well-being of individuals and communities (Desmet and Pohlmeier, 2013, p. 6). Positive design can be placed within positive psychology. Positive design aims to design for pleasure (personal affect), personal significance (pursuing personal goals), and virtue (being a morally good person). All three should be strived for in each design. Such designs are then held to enhance overall well-being (van den Hoven et. al., 2015, p. 377). - According to Buwert (2017) “<i>Potentiality</i> is the condition under which the ethical appears. Finding oneself existing in a mode of potentiality – being equally as able to do as to not do something – the issue which arises is: which option to choose? Once it is possible for me to say both that “I can” and “I can-not” do something, which should I? On what basis do we make decisions to enact our potentiality or nonpotentiality (to act or to act by not acting) in response to an encountered situation?” (Buwert, 2017, p. S4461).	Gray et. al. (2022) McCollam (2014) Desmet and Pohlmeier (2013) Van den Hoven et. al. (2015) Ahmed (2020) Buwert (2017)
1.4 Design Cognition (27) - Hermeneutics - Philosophy of design (Duty and Virtue ethics) - Reflective practice	- <i>Hermeneutics</i> is the study of interpretation. Hermeneutics plays a role in several disciplines whose subject matter demands interpretative approaches, characteristically, because the disciplinary subject matter concerns the meaning of human intentions, beliefs, and actions, or the meaning of human experience as it is preserved in the arts and literature, historical testimony, and other artifacts. (George, 2020) - <i>Design philosophy</i>, the philosophy of design is a roadmap, a guiding principle that steers the creative process. It deals with core areas as the designers’ values, beliefs, and goals and is the “why” behind the “what” and the “how” (Daisie, 2023). - <i>Reflective practices</i> are, at the very core, being able to reflect on one’s own actions to then engage in a process of learning (Visser, 2010).	1.4 Pérez-Gómez (1999) Nocek (2022) Newton (2004) Nunes (2017) D’Anjou (2011)
2. Design Education (18) - Design pedagogy - Educating ethical designers - Legitimation Code Theory (LCT)	- Lowyck (2002) defines <i>Pedagogical design</i> as: “(..)a novel term in the conceptual landscape of instruction, teaching, learning and learning support, refers to any systematic choice and use of procedures, methods, prescriptions, and devices in order to bring about effective, efficient, and productive learning (e.g., Romiszowski, 1981).” (Lowyck, 2002, p. 199) - <i>Legitimation Code Theory (LCT)</i> is a framework for analyzing and shaping knowledge practices to advance social justice and knowledge-building. LTC is being used to both interpret the world in various ways and to change it. Concepts from the framework reveal different dimensions of what Bourdieu called “the rules of the game” which is the bases of achievement underlying social fields of practice. These are often unwritten/unspoken they “go without	2. Clune (2011) Das et. al. (2023) Giloi and Quinn (2019)

	saying” in ways that, when accessible only to actors from specific backgrounds, generate social inequality. By making such organizing principles visible, LCT enables these bases of achievements to become accessible to more actors and in this manner, is promoting social justice (Maton et. al., 2015, p. 3; Giloi and Quinn, 2019, p. 833).	
3. <u>Participatory/Co-creation</u>(31) - Collaboration - Innovation - Open design - Workshops - Situated ethics.	- <i>Open design</i> can be defined as the internet-enabled collaborative creation of artefacts by a dispersed group of otherwise unrelated individuals. Creatively, open design promotes the unprecedented sharing of knowledge between the professional and the amateur designer, breaking down potential unnecessary barriers. When carried out for the common good rather than for commercial profit, open design allows the sharing of creative skills between developed and undeveloped nations for humanitarian benefit, which is countering the more negative ramifications of global product consumerism (Cruickshank and Atkinson, 2014, p.3).	3. Almendra and Falcão (2017) Ross (2010) Eger and Drukker (2012) Cruickshank and Atkinson (2014) Cooper (2024) Frauenberger et. al. (2017) Jönsson et. al. (2005)
4. <u>Responsibility/Re-sponsible design</u>(15) - CSR Corporate Social Responsibility - Environmental Re-sponsible design ERD - Responsible innovation - Responsibility of the designer	- Ahmed (2020) defines <i>responsible design</i> as “(..)a collective effort to influence graphic designers to make informed choices regarding the ethics and larger societal consequences of their design work – is a growing, reformative social movement (Aberle 1966).” (Ahmed, 2020, p. 32) - <i>Corporate social responsibility (CSR)</i> has become an increasingly important aspect of business and business research within recent years. It refers to the responsibility of companies to contribute to sustainable development by balancing areas such as economic, social, and environmental concerns (Koo, 2016, pp. 49-51). - <i>Environmentally responsible design (ERD)</i> is a design approach that considers the environmental, economic, and social implications of design decisions throughout the life cycle of a service, product, or system (LI, 2024).	4. Ahmed (2020) Boehm (2015) Bentham (2017) Koo (2016)/ Koo and Cooper (2016) Melles et. al. (2011) Rivard et. al. (2021)
5. <u>Social Design</u> (21) - Design for social Innovation - Socially responsible design SRD - Design justice /activism	- “<i>Design for Social Innovation</i> uses design to generate equitable and sustainable outcomes for people and the world we live in. In this subject, we identify and define problems worth solving—complex social, environmental, and indigenous worldview issues that have both local and global impacts. We then design solutions using human-centered, society-centered, and planet-centered research methods and a toolkit of innovative design practices” (V., University, 2021). - <i>Socially Responsible Design SRD</i> is the use of design to address issues of the environmental, economic, and social kind. SRD focus on moving beyond first world consumer demands towards an approach to product design that is more holistic and responsible and able to embrace cultural, ethical, and humanitarian values. In other words, ERD is 'design for good' / 'design for need' (Silva and Simones, 2010, Ch. 10). - <i>Design justice</i> “is an approach to design that is led by marginalized communities and that aims explicitly to challenge, rather than reproduce, structural inequalities. It has emerged from a growing	5. Cipolla and Bartholo (2014) Das et. al. (2023) Koo and Cooper (2016) Tromp and Vial (2023)

	community of designers in various fields who work closely with social movements and community-based organizations around the world.” (MIT Press, 2020) - <i>Design activism</i> represents design central role in areas such as: promoting social change, raising awareness about values and beliefs (e.g., in relation to areas such as sustainability and environmental changes), and questioning the constraints that for example mass production/consumerism place on the everyday lives of people (Markussen, 2013, p. 38)	
6. <u>Sustainability</u> (43) - Sustainable behavior - Sustainable technology - Product life cycle - Sustainable design Theory/Methods - Ecological design	- <i>Sustainable/sustainability defined in Merriam-Webster as:</i> “relating to or being a method of harvesting or using a resource so that the resource is not depleted or permanently damaged” (Merriam-Webster 2, 2024). Sustainability in design is about Designers and manufacturing decision makers who adopt a sustainability focus and establish a sustainability culture within companies are more likely to be successful in enhancing design and manufacturing (Rosen and Kishawy, 2012, p. 154). - <i>A product's life cycle</i> talks about the ethical behavior of a company. It concerns how products are managed in terms of marketing schedules, expansion to new markets, prices type of packaging and package re-designs e.g., moving from plastic to bio-degradable material. Such a lifecycle is normally divided in four stages which concerns introduction, growth, maturity, and decline of any product (Eger and Drukker, 2012).	6. Bhamra et. al. (2011) Wilson et. al. (2016) Findeli (1994) Eger and Drukker (2012) Benson and Napier (2012) Sherwin (2004) Salazar and Baxter (2015)
7. <u>Technology</u> (35) - AI - Futuring - Collingridge Dilemma - Speculative design - SynBio/biomimetics - Persuasive & user influencing technology. - Techno ethics - Value sensitive design (VSD)	- <i>The Collingridge dilemma</i>, states that early in the process of development of a technology, there will be significant possibilities to intervene in the process, however, information about its unintended or undesired outcomes is scarce, while later on in the development of the technology, because of the technological momentum, “steering of technology is either impossible, marginally possible, or prohibitively expensive (Collingridge 1980)” (van den Hoven, 2015, p. 2) - <i>Persuasive design/Technology</i> is concerned with influencing user behavior through design decisions. Examples of persuasive technology include speed bumps that force a driver to slow down and windows that open automatically to regulate the temperature within the building (Wilson et. al. 2016, p. 89). - <i>Value Sensitive Design (VSD)</i>: is a concept that promotes the consideration of human principles and standards in the planning of technology. Bringing human values into the forefront of the technical design process, VSD provides technologists, designers, business leaders and other stakeholders with strategies for identifying and . Value-sensitive design aims to consider the values not only of the users, but of all others impacted by the technologies - regardless of whether those individuals will ever actually use the technologies (Pratt, TechTarget, 2024).	7. Banks et. al. (2019) Lomas and Xue (2022) Cooper (2024) Zhang (2022) Haidamous (2017) Knowles and Davis (2016) Shilton and Anderson (2016) Van den Hoven (2015)
8. <u>Human-centered design</u> (14) - HCI Human computer interaction - Inclusive design - Individualisation - Socio technical systems STS	- <i>Human-centered Design (HCD)</i>: The HCD philosophy incorporates ethnographic field research and qualitative interviews to empathize with users and understand their needs (Forlando, 2016, p. 44). - The practice behind designing for <i>inclusivity</i>, i.e., designing so that as many people as possible can use the product or service being developed, is known as <i>inclusive design</i>. Within inclusive design is an awareness of the pervasiveness and importance of human diversity	8. De Bont (2021) Forlando (2016) Frauenberger et. al. (2017) Heyligen and Dong (2019) Keionen (2010)

- UX user experience design	and the injustice to which neglecting this may lead (Van den Hoven, 2015, p. 241,385). - <i>STS, a socio-technical-system</i> is a collection of devices, people, and procedures for collecting, storing, processing, retrieving, and disseminating information, for a particular purpose (van den Hoven, 2015, p. 316).	
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4.2 Main findings and highlights

As reflected in chapter 3.1, a diverse set of themes has been identified from our dataset. Based on the data-analysis the following 8 central themes were established in literature:

Table 4: All 8 themes in a 5-year span evolution.

1.	Design processes and practices - <i>Decision making (Agency, intentionality, Rational choice theory, consequences)</i>	(48) (21)
2.	Sustainability - <i>Sustainable behavior (situated ethics, ethical guidelines, morality, harmony/balance)</i>	(43) (16)
3.	Technology - <i>AI, emergence, futuring, system design, value-sensitive design.</i>	(35) (18)
4.	Participatory/Co-creation - <i>Collaboration, innovation, situated/responsive ethics.</i>	(31)
5.	Social design - <i>Socially responsible design</i>	(21)
6.	Design education - <i>Design pedagogy, developing ethical character, Legitimation Code Theory (LCT)</i>	(18)
7.	Responsible design - <i>Environmental responsible design (ERD), Corporate social responsibility</i>	(15)
8.	Human-Centered design - <i>(STS) Socio-technical systems, user experience.</i>	(14)

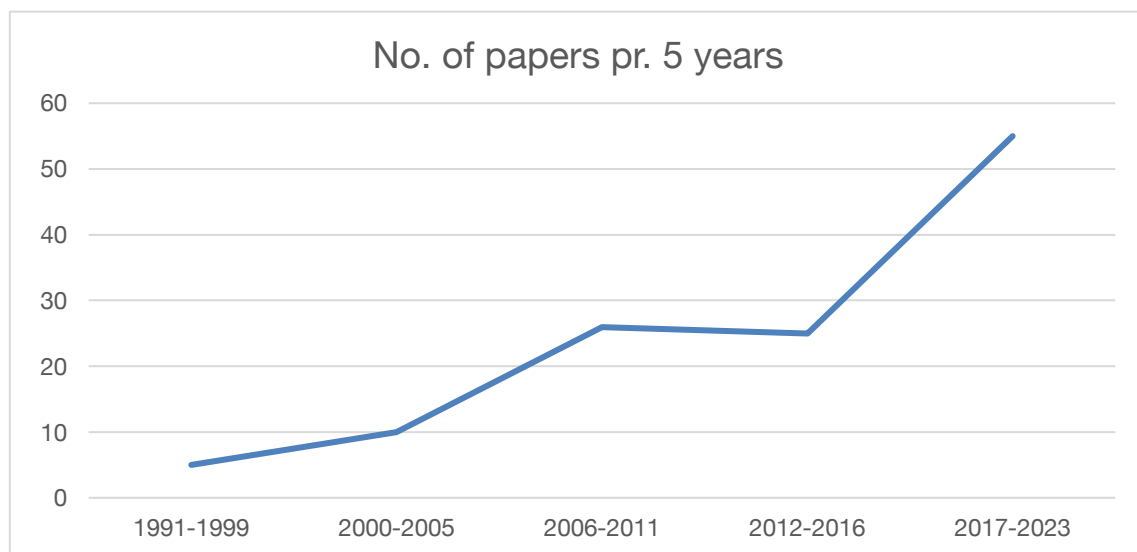


Figure 1 All 8 themes in a 5-year span evolution.

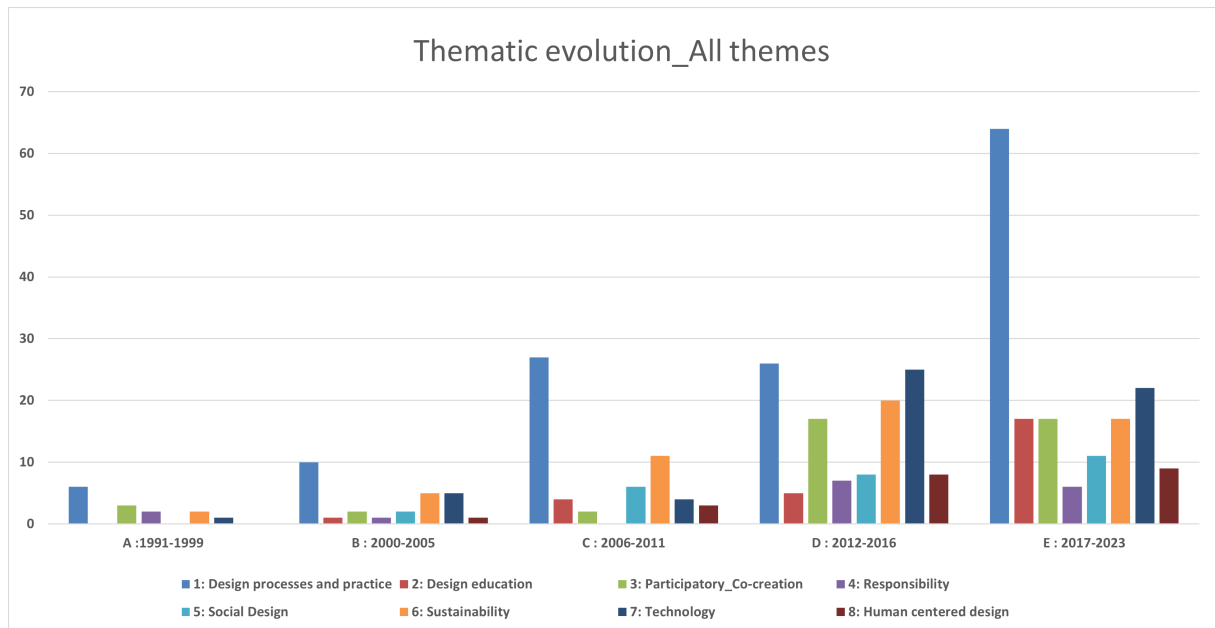


Figure 2. All 8 themes in a 5-year span evolution.

Where the three predominant themes are related to questions of (see Figure 2): the designers' role and actions (Design processes and practices), the effects and consequences of new developments (Technology), and the common objectives of addressing the urgency of ecological imbalance (sustainability).

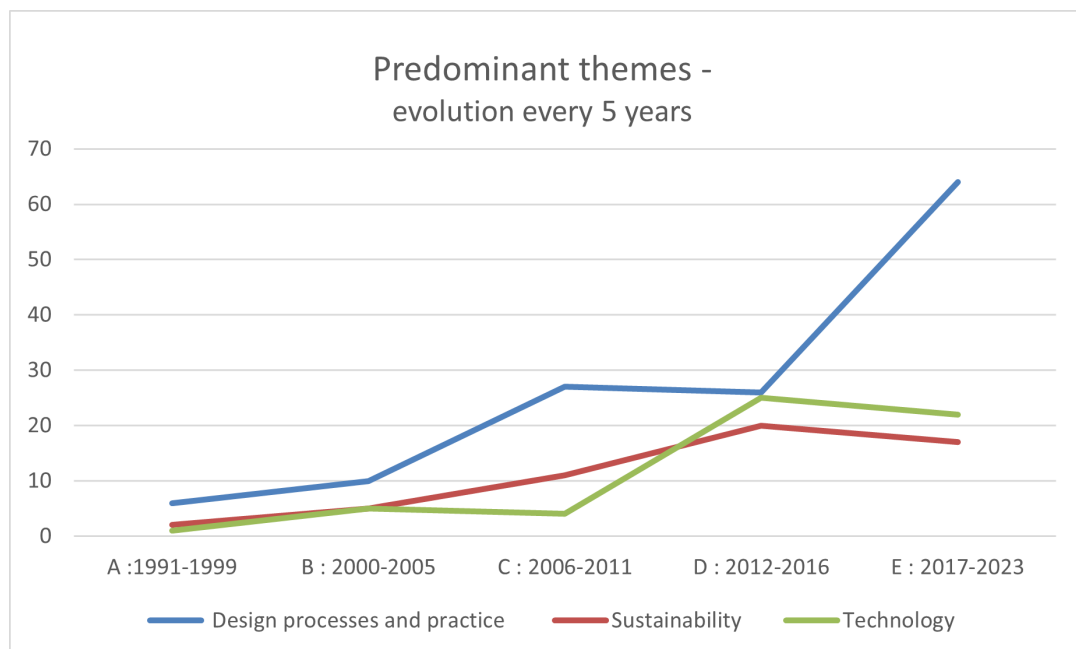


Figure 3. The three predominant themes (Design processes and practice, Technology, and Sustainability)

4.2.1 Shortcomings and limitations

Data: The topic of design processes and practice is somewhat broad, as shown in Table 2 this node contains broader sub-themed nodes (decision-making, design cognition, design methods and tools, and thematic frameworks). Therefore, a natural incline would be expected in comparison to the other main themes.

Data-search: The con of doing a search maximum of five journals is that it is more time-consuming. However, if we do one large cluster search, we will have a poor chance of troubleshooting any inconsistencies or errors as fast and precisely, as when focusing on a select few journals at a time. This is one of the limitations, and time-consuming factors, that follow in searching specific sources (pre-listed journals) as well as a specific type of data source (journal articles).

5. Discussion

In this study, a structured literature review has been conducted to answer the following research question: Which topics are commonly researched in design ethics literature and how has it evolved? If we look at the thematic evolution in the investigated journals the data analysis revealed a clearly defined and significant increase in the concept of ethics in general spanning over the last 7-8 years. The increase has primarily been driven by three overall themes; firstly, the design processes and practices, second the technology and third the sustainability. They represent three different types of drivers for ethical considerations: the role, the object, and the consequences of design.

5.1 Design processes and practice: The role of design and designer

The role of design and the designer has received the most attention. Under the theme of Design processes and practices, a topic such as decision-making is predominant. The discussions have specifically revolved around 'Agency' and 'Intentionality', thus focusing on the role of the designers and the consequences of the choices made by the designer. Various methodologies already exist to infuse values and ethics into the design process, such as value sensitive design (Friedman & Hendry, 2019). The education of the designer has also increased significantly over the last decade, just adding to the emphasis on the designer. This indicates a growing interest from researchers to investigate this role.

But interestingly other topics concerning the role of design and designer have received comparatively less interest. The engagement with society and users (Participatory Design) and the accountability towards society and the environment (Responsible Design) seem to evade lost interest. Is this shift in interest towards the designer's role, processes and practices a result of increasing awareness of what impact on the surrounding world the designers are part of creating? But why does this not boost the interest in e.g. Responsible Design? Is it easier and less controversial to explore the agency, philosophies, and practices, rather than e.g. Responsible Design, that may intrinsically have more accountability associated with the theme? Are we shying away from topics that may sound more activist and require a stance in a complex and paradoxical problem space?

5.2 Technology: The object of design

What we design is intrinsically part of the processes and practices, but the single theme that has emerged as an ethical subject is technology. This theme has received attention from researchers, in terms of how its design and technologies co-evolve over time (Laursen and Barros, 2022). However, it seems to level out recently which is somewhat surprising considering the latest technological development within AI and IoT (Internet of Things). It has a profound effect on the objects we design and develop, to the extent that it becomes unpredictable what the unintended applications could be (Collingridge dilemma). Given this impact, it is puzzling that the attention is not higher. And if we take the latest versions of online AI services into account, the ability of technology to generate visuals of objects and worlds (Midjourney) and content for videos this seems even more crucial to address. Does technology now transcend from object-related consequences for the users and society to a role-challenging entity? With AI imposing on creativity, does that not more than ever call for emphasis on the ethical consideration related to the creation of “the new”? Do we need a manifest of what designers and creators of new things need to consider, or processes to include (participatory?) when creating something new?

5.3 Sustainability: The consequences

Climate crisis, environmental changes, and the realization amongst humans that the resources of the planet are finite has promoted the theme of Sustainability for many years. “One paradox of our times seems to be our wish to establish a sustainable society on the basis of market principles.” (Jeslma, 2006; p 221). The relation between using resources for products, and their negative impact on the environment is no longer challenged. However, we still have unsolved issues concerning how we deal with this as designers. The complexity of the topic on an operational level is immense (Lund et al., 2023), e.g. how do evaluate the environmental consequences of a product based on a conceptual drawing?

Literature highlights ethical concerns surrounding planned obsolescence and the longevity of products (Cooper, 2008; Frahm et al. 2022), the issues of ‘repairability’ (Nielsen et al., 2023), and the imperative for a shift towards designing upon resource scarcity (Feast and Laursen, 2023). When we need to deal with sustainability issues, use fewer resources and perhaps generate fewer new solutions, then how will this affect societies globally? Will sustainable solutions be available to the masses, to the developing world? Interestingly the theme of sustainability is levelling out, and themes like Social Design concerned with inclusivity have received little attention from design researchers. Surely there is a need to understand how the needed changes in our consumption of resources and transition “to green” will affect societies all over the world. If we want global impact on the environment, then how do we get global buy-in to the change? This calls for understanding social design aspects such as social justice and social responsibility.

6. Conclusion: What are we missing?

Combining the three main themes identified, several aspects are missing. In this era of the need for transformation, information overload and change, we need to be mindful of the areas of research we are neglecting. There is an abundance of systemic, incomprehensible interdependencies and Collingridge dilemma dimensions regarding the consequences of any choice made. However, there is a lack of clarity regarding consequences and the directions to take. In the search for 'what is good' or 'virtuous', how are we to move forward? We call for more empirical studies, on what are we missing. We encourage more research on the future of design ethics, and the underlying presumptions in current research. The wicked problem, the overload of information, endless choices and analysis of consequences leaves us either paralyzed or perplexed. Any large-scale behaviour change requires a combination of focus in the action and clarity in the path towards the vision. How can we attain focus and priorities, when researching ethics in design?

How may we for instance collaborate? How may different actors be responsible? As researchers, we could for example build a joint design ethics etiquette that applies to design, with and without AI technology, that helps facilitate the transition of the field of design to reach a sustainable equilibrium within the planetary boundaries for viable businesses and human well-being. How may we make design ethics operational, accessible, understandable, and collective? Can we develop a code of conduct? We suggest design ethics educators and scholars go together and collectively take on a responsibility. It could be in raising research discussions, joint agendas, or operational codes of conduct for 21st-century design activities. The doctor oath is one example of a profession going together, agreeing on ethical concerns for a profession. How may one such look in design?

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